

Dream Preconstruction Workflow

The Missing Intelligence Layer in Preconstruction

Problem Statement & Strategic Opportunity

Opening Problem Statement

Preconstruction teams operate at the center of enormous complexity. Teams must translate drawings, specifications, and subcontractor proposals into structured takeoff and cost data across multiple disconnected systems. Despite modern cloud-based estimating and web-based takeoff tools, the starting point for nearly every estimate remains the same: a blank canvas.

Today's workflows require estimators to manually decide what to measure, how to measure it, how scope should be assembled, and how it maps to company standards. This setup work is repeated on every project, is highly dependent on individual experience, and introduces inconsistency across teams.

The industry has focused heavily on automating measurement. The larger and more valuable opportunity lies earlier in the process: automatically generating the takeoff framework itself.

Preconstruction is missing an intelligence layer that converts documents into structured, estimator-ready scope and assemblies before measurement begins.

This gap represents one of the largest remaining friction points in the preconstruction workflow and one of the biggest opportunities for step-change productivity gains in preconstruction.

Vision: Preconstruction Avengers – Origin Story



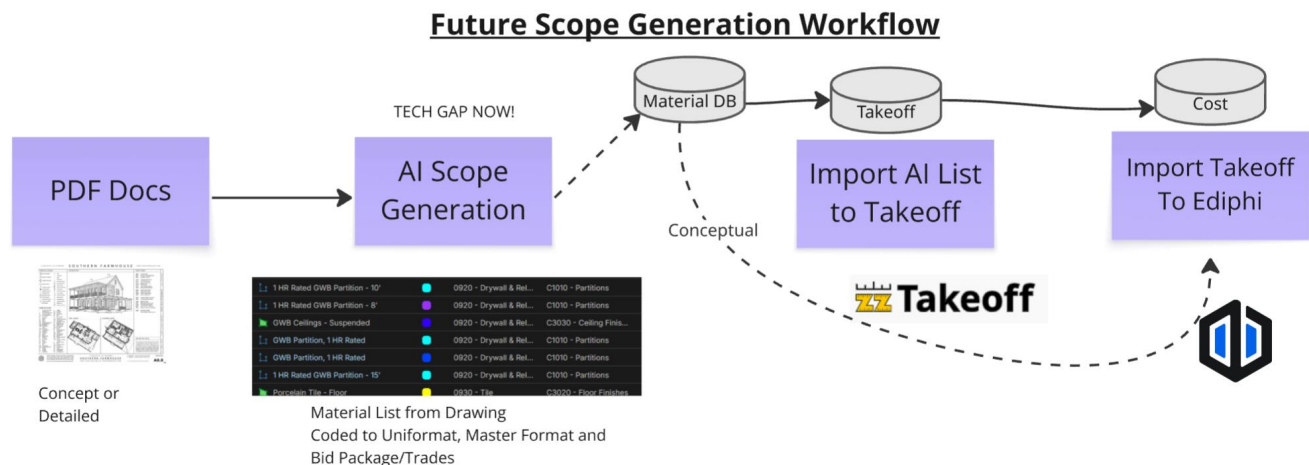
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Core Concept

Create an AI-driven scope generation layer that sits between PDFs, takeoff platforms, estimating systems, and subcontractor proposals.

This layer prepares structured takeoff conditions, assemblies, and predictive scope so estimators can focus on measurement and judgment, not setup.

Future Scope Generation Workflow



The Three Data Worlds of Preconstruction

Successful scope generation must operate across three parallel classification systems:

- Uniformat – **Where** the object is located
- MasterFormat – **What** the object is
- Trade / Bid Packages – **How** the work is purchased

A viable AI model must understand and translate between all three.



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Hard Bid Workflow (Complete Documents)

Input:

- Drawings & Specs

Output:

- Auto-generated takeoff conditions (Basically template setup for YOUR job)
- Parent-child assemblies
- Measurement types
- Default assumptions

Estimator Action:

- Import into zzTakeoff
- Perform measurement
- Push quantities to Ediphi

Outcome:

- Faster setup
- Consistent assemblies
- Cleaner material rollups

Conceptual Workflow (Sparse Documents)

Input:

- Concept sketches
- Narratives
- Early geometry

Output:

- Measurable geometry to takeoff
- Predictive scope to estimating

Outcome:

- Hybrid model: measured where possible, predicted where necessary

Why Subcontractor Proposals Matter

Subcontractor proposals reveal how scope is packaged, interpreted, and bought. They provide high-quality training data that links drawings, takeoff, and estimates to real-world pricing behavior.

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Bid leveling is only the entry point. The deeper value is model training from the below.



Why This Is Strategic

This intelligence layer does not replace takeoff or estimating platforms.

It orchestrates them.

Whoever owns this layer becomes the system of intelligence for preconstruction.